HOMOLOGY AND MAYER–VIETORIS ON THE SEMI-SEGAL FOUNDATION

We prove homology results conditional on a category D with finite semi-Segal structure, copowers, and stability. Semi-Segal structure alone does not give homology. This note separates the assumptions from the constructions and theorems checked in `sag/hcat-segal/All.seg`.

1. The index type and the selected structure

The code assumes an indexing type named `Cat`. For each index `C:Cat`, it supplies a type `objectsC`, arrow types `arrowsCab`, and the finite semi-Segal data described below. It then selects `D:Cat` and equips this particular structure with the additional assumptions for homology. Throughout the note, D denotes this selected structure, and $D(a, b)$ means `arrowsDab`.

The name `Cat` is inherited from the original development. Here it is only an index type: no types of functors between its indices, or ambient categorical operations on it, are supplied. It is not asserted to classify all categories. The indexing arrangement allows reuse of the existing theorem files; for the homology results one could instead assume a single finite semi-Segal structure D directly.

The results therefore concern the internal structure of D. They do not construct the infinitely coherent ambient category of categories considered in the original project, nor derive these internal semi-Segal inputs from its axioms.

2. The categorical axioms and their consequences

We assume a type of objects and, for each pair a, b, a type of arrows $D(a, b)$. The arrow types need not be sets. We supply simplex data through dimension four, subject to the following filling conditions.

(1) *Triangles.* For composable arrows $f : a \to b$ and $g : b \to c$, there is a family $A_2(f, g, h)$ of triangles, indexed by arrows $h : a \to c$, and the type
$$\sum_{h:D(a,c)} A_2(f, g, h)$$
is contractible. We *define* $g \circ f$ to be the third edge of the chosen centre of this type.

(2) *Tetrahedra.* For a compatible tetrahedral boundary with one specified triangular face missing, the type of completions (the missing triangle together with a tetrahedron) is contractible. Applied to a boundary made from the chosen composites, this *constructs* the associativity path.

(3) *Four-dimensional simplices.* For a compatible boundary with one specified tetrahedral face missing, the type of completions (the missing tetrahedron together with a four-dimensional simplex) is contractible. A particular such filling *proves* the pentagon for the associativity path just constructed.

Contractibility means a chosen completion together with a path from it to every completion. It retains higher coherence, rather than merely asserting existence or uniqueness in a set.

Precisely, the implementation uses the inner horns Λ^n_1 for $n = 2, 3, 4$. Their missing faces are respectively 02, 023, and 0234. The boundaries are normalized in stages: triangle data become paths from selected composites, and tetrahedron data become equalities of triangular pastings. The required normalization equivalences are proved. We do not assume or construct an infinite semisimplicial object here.

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We also assume identity arrows and two families of triangles on $(1_a, f, f)$ and $(f, 1_b, f)$. The unit laws follow from these triangles. Thus composition, associativity, the unit laws, and `hpent` are derived; they are not separately supplied equations. All laws are paths in ordinary dependent type theory, with no judgmental categorical laws.

3. Additional assumptions for homology

For one such D, we assume the following data and universal properties.

(1) *Copowers by small types.* For every small type X and object A, an object $X \otimes A$ and arrows $\eta_x : A \to X \otimes A$ such that restriction is an equivalence:

$$D(X \otimes A, T) \longrightarrow \prod_{x:X} D(A, T), \qquad q \longmapsto (q \circ \eta_x)_{x:X}.$$

This is a universal property of the supplied maps, not just an arbitrary equivalence between these types.

(2) *A zero object* 0. Both $D(0, T)$ and $D(T, 0)$ are contractible for every T.

(3) *Suspensions.* For every B, a chosen pushout $\Sigma B = 0 \amalg_B 0$, including its square and universal property.

(4) *Stability.* Every pushout square in D is also a pullback square. This is stated representably: applying $D(S, -)$ gives a pullback of types for every S.

(5) *A distinguished object* G. The code calls it `unitD`. No monoidal structure or monoidal-unit property is assumed.

The foundations are ordinary dependent type theory with universes and the existing `Theory` parameters `univ`, `images`, and `surjections`: univalence and replacement, expressed by small image factorizations and their surjectivity. These supply the truncations and sets of components used below. Function extensionality is derived. No new axioms beyond these foundations and the explicit categorical and homology inputs are used in the results below.

4. What is proved: homology and its maps

For a coefficient object A of D, we *define*

$$H_n(X; A) = \pi_0 \mathsf{Hom}_D(\Sigma^n G, X \otimes A), \qquad n \geq 0.$$

Here $X \otimes A$ is the copower indexed by the type X. The development constructs abelian groups on these sets, including H_0. It proves functoriality of the induced maps. Multiplicativity in positive degrees is proved in `HGroupMaps.seg`, and in degree zero in `StableGroupMaps.seg`. The file `HomologyMaps.seg` proves that homotopic functions induce equal maps and that equivalences of types induce equivalences on every H_n; these are now group isomorphisms in every nonnegative degree. The equivalence proof applies π_0 to postcomposition with the invertible tensor map. Commutative squares of functions induce commutative squares on homology by functoriality and homotopy invariance.

The new file `HomologyUnits.seg` identifies every chosen group unit with the class of the categorical zero map. In positive degrees this uses the existing pointed suspension–loop equivalence. In degree zero, the cone of zero is $(0, 0, p \cdot p^{-1})$. Cancelling this loop and then the two contraction paths shows that the stable loop equivalence sends zero to the constant loop. Transport of the loop-group unit gives the claim. Postcomposition preserves the zero class, so the induced maps preserve units in every degree.

The degree-zero additivity proof uses the existing stable equivalence

$$\sigma_{B,E} : D(B, E) \simeq \Omega D(B, \Sigma E).$$

Write $L_{E,T} : D(\Sigma E, T) \simeq \Omega D(E, T)$ for the suspension universal-property equivalence. Define the suspension of an arrow $f : E \to T$ by

$$\Sigma f = L^{-1}_{E,\Sigma T}(\sigma_{E,T}(f)).$$

The theorem `StableNaturality.sigma-natural` identifies $\sigma_{B,T}(f \circ c)$ with the pointed loop map of postcomposition by Σf applied to $\sigma_{B,E}(c)$. Its proof normalizes $\sigma(c)$ as precomposition of the suspension meridian by c, and $L(g)$ as postcomposition of the meridian by g. Associativity compares the two orders of composition. Commutativity of loops in stable mapping types makes the comparison independent of the chosen pointing paths. On components this is a square of equivalences with a loop-group homomorphism, proving additivity for the original degree-zero group structure. No new axioms are used.

5. What is proved: excision and Mayer–Vietoris

Excision is a theorem: the copower $(-) \otimes A$ sends a given pushout square of small types to a pushout square in D. The proof uses the copower universal property and the derived pentagon. Stability then makes the resulting square a pullback as well, giving the fibre descriptions used for exactness. Existence of all pushouts of small types is not an additional premise of this conditional theorem: the particular pushout and its universal property are supplied.

Let X be the pushout of $B \xleftarrow{i} C \xrightarrow{j} E$, with maps $u : B \to X$ and $v : E \to X$. Suppress A in the notation. The unchanged `HomologySequence.seg` proves, for every $n \geq 1$,

$$H_{n+1}B \oplus H_{n+1}E \longrightarrow H_{n+1}X \xrightarrow{\partial} H_nC \longrightarrow H_nB \oplus H_nE \longrightarrow H_nX.$$

All arrows are homomorphisms and the three interior positions are exact. The non-boundary maps are (i_*, j_*) and $u_* - v_*$. The structural boundary is the same map in both exactness assertions; adjacent degree blocks have the same difference map.

The new `MayerVietorisTail.seg`, together with `MayerVietorisZeroGroups.seg`, joins this sequence to

$$H_1B \oplus H_1E \longrightarrow H_1X \xrightarrow{\partial_1} H_0C \longrightarrow H_0B \oplus H_0E \xrightarrow{u_*-v_*} H_0X.$$

All arrows are homomorphisms, including the existing connecting map ∂_1, and all three interior positions are exact. The equalizer statement below is equivalent to exactness at the product by group cancellation. The map ∂_1 is the direct boundary with source object G. Its incoming difference is definitionally the degree-one difference above. The existing cone and fibre exactness theorems give

$$\operatorname{im}(u_* - v_*) = \partial_1^{-1}(0), \qquad \operatorname{im} \partial_1 = (i_*, j_*)^{-1}(0, 0).$$

The image of (i_*, j_*) is the equalizer of the two final maps $u_* \operatorname{pr}_1$ and $v_* \operatorname{pr}_2$. Images mean merely inhabited fibres, expressed by propositional truncation; no choice of preimages is asserted.

6. Why the final boundary is additive

Work first with a pushout square $B \leftarrow C \to E$ with apex P in D, and a test object S. The two zero arrows $B, E \to \Sigma C$, together with the suspension meridian of C and the zero-map identifications, form a cocone. The pushout universal property constructs an arrow $\beta : P \to \Sigma C$ with that cocone. This construction is `ConnectingArrow.Arrow.connecting`.

Let $\delta : D(\Sigma S, P) \to D(S, C)$ be the *existing* direct boundary. The comparison theorem proves

$$\sigma_{S,C}(\delta(g)) = \Omega(\beta_*)\big(L_{S,P}(g)\big), \qquad g : D(\Sigma S, P),$$

where β_* is postcomposition with β, equipped with a specified path identifying its value at zero with zero. To prove the comparison, evaluate each pullback cone against the cocone of β. The pentagon compares the two orders of composition; coherence of precomposition with zero identifies

the endpoints. At the constant loop, this evaluation is constant. Consequently its two endpoint paths agree, so on arbitrary loops the evaluation is precisely a pointed loop map.

On components, L and σ are the equivalences used to define the original source and target group structures. A pointed loop map preserves loop concatenation, hence transporting its homomorphism law along these same equivalences proves

$$\partial_1(x + y) = \partial_1(x) + \partial_1(y).$$

This is `BoundaryAdditivity.Types.hom`, also exposed as `MayerVietorisZeroGroups.Types.delta-hom` for the exact tail above. There are no additional axioms or diagram-compatibility hypotheses. All coherence here follows from identity types, the derived pentagon, and the given universal properties.

7. Boundary naturality: the precise proved statement

Consider two pushout squares in D, vertical arrows a_C, a_B, a_E, a_P, and homotopies $u_1 a_B \simeq a_P u$ and $v_1 a_E \simeq a_P v$. For a fixed test object S, these homotopies induce a map N_S between the types of cones on the two cospans after applying $D(S, -)$. A *coherent map of representable squares*, in the theorem as implemented, additionally supplies the equality

$$\mathrm{cone}_1(a_C \circ c) = N_S(\mathrm{cone}(c)), \qquad c : D(S, C).$$

This is the specified cube compatibility for the diagram map. It is an input to the naturality theorem, not a new axiom about D, and does not mention a boundary map.

From this data, `MayerVietorisNaturality.seg` proves

$$\partial_1 \, (a_P)_* = (a_C)_* \, \partial$$

for the direct boundary on components, including the final boundary into degree zero. The two pointing compatibilities needed for this argument are proved from the existing zero-map coherence laws.

`BoundaryLoopComparison.seg` proves that the structural boundary used in the positive-degree exact sequence equals the loop of the direct boundary, transported through the chosen suspension equivalences. Consequently `BoundaryNaturality.seg` proves the corresponding naturality statement for that structural boundary. Its input is the same coherent map of representable squares. No comparison with a separately chosen direct boundary in the next degree is asserted.

8. What remains open

The naturality theorems above require the stated cube compatibility; constructing this compatibility from maps of pushout diagrams of types is still required for a natural long exact sequence on those diagrams. There is no terminal surjectivity claim onto $H_0 X$. These are conditional, generalized homology results. We have not constructed a model of the assumptions or identified this construction with ordinary homology with specified coefficients. Model construction is independent work.

The existing additivity and reduced suspension results are retained. The separate sphere calculations additionally assume suspensions of small types with their pushout universal properties, and a choice of desuspension in D with paths $\Sigma(\mathrm{desusp}\, B) = B$. Those inputs are not needed for the Mayer–Vietoris theorem above.

9. A concrete interpretation to construct

The intended external example is the stable higher category of spectra, with G the sphere spectrum. The copower is $X \otimes E = \Sigma^\infty_+ X \wedge E$, so the displayed definition of homology becomes

$$H_n(X; E) = \pi_n(\Sigma^\infty_+ X \wedge E).$$

Taking the Eilenberg–MacLane spectrum $EM(M)$ of an abelian group M should identify this with ordinary unreduced homology with coefficients in M. This is the standard external interpretation, not a checked interpretation of our axiom system; see Lurie, *Higher Algebra*, §1.4, and Nardin, *Introduction to stable homotopy theory*, Chapter 2, especially Exercise 20 for the ordinary-homology comparison.

There is now a checked first ingredient in `sag/hcat-spectrum/EMSpectrum.seg`. Reusing `sag/em`, it constructs pointed types

$$E_0 = M, \qquad E_{n+1} = K(M, n+1), \qquad \Omega E_{n+1} \simeq_* E_n.$$

At the first stage the equivalence is the band of M-torsors, and its preservation of addition is checked. At later stages it is the existing delooping equivalence. These are coefficient Ω-spectrum data; they do not yet define the higher category of spectra or its copowers.

The remaining interpretation work is to construct mapping types and the finite semi-Segal structure for spectra, establish the copower and stability properties with the required universe sizes, and prove the comparison with ordinary homology, including its maps. The presheaf, fibration, universe and univalence constructions in `unit/core` provide semantic infrastructure. They do not by themselves supply these spectral constructions or a verification of every parameter of this homology root.

References

[1] J. Lurie, *Higher Algebra*, 2017. `https://www.math.ias.edu/~lurie/papers/HA.pdf`.

[2] D. Nardin, *Introduction to stable homotopy theory.* `https://ncatlab.org/nlab/files/Nardin-StableHomotopyTheory.pdf`.

10. Validation and preservation

The homology root checks with 2876 declarations and no failures in a fresh `--opened` run (3.13 seconds). The coefficient-spectrum root checks with 1001 declarations and no failures (0.14 seconds). The original 169 `hcat` source files are preserved; shared theorem files are relative links. The continuation uses the horn-derived pentagon. It does not identify the new groups with groups formed using the old chosen categorical operations.

11. Why this may be interesting

The main point is that a finite amount of categorical coherence suffices for the homology results proved here. Simplex data through dimension four supply composition, associativity and their pentagon; together with the stated universal properties, they support abelian groups, excision and Mayer–Vietoris in every nonnegative degree. Iterating suspension increases the homological degree without requiring an infinite semisimplicial object in the formalization. This establishes sufficiency of the given finite data, not their minimality.

The additivity of the final boundary makes this more than a construction of groups and an exact sequence of pointed sets. The boundary defined by the pullback property is proved compatible with the group structures defined by suspension. Its comparison with a pointed loop map explains why these independently defined constructions agree. The coherence needed for that agreement is proved in ordinary dependent type theory, using the pentagon derived from a horn filler.

The development also separates two mathematical tasks. A model must supply the finite semi-Segal data and the copower, zero, suspension and stability properties; the homology proofs can then be reused. These model obligations are substantial, but they are explicit. The chosen object G specifies which maps detect homology, so different choices of D, G and coefficients may give different theories. A verified spectra model, with G the sphere spectrum and Eilenberg–MacLane coefficients, would connect this finite foundation to ordinary homology. That interpretation and the remaining

naturality comparison are still needed. The present contribution is the checked conditional argument and an explicit account of the coherence it uses.